

ANSWER KEY

Name: _____

Roll No.: _____

AMRITA VISHWA VIDYAPEETHAM
Amrita School of Computing, Bengaluru Campus
Department of Computer Science & Engineering
B.Tech CSE Third Semester (2025-26)

23ECE205 DIGITAL ELECTRONICS
QUIZ 01 (SET A)

Duration: 15 minutes

Date: 01/08/2025

Max. Marks: 10

CO	Course Outcomes
CO1	Realize a given expression in terms of basic building blocks.
CO2	Minimize a given logic expression.
CO3	Design combinational circuits
CO4	Design Sequential circuits.

Q. No.	01	02	03	04	05	06	07	08	09	10
Marks Obtained										
Total (out of 10)				Sign of student		Sign of faculty				

Q1. Write Truth table of $f(a, b)$ given, output is high if "input is odd and greater than one".

Ans:

a	b	$f(a, b)$
0	0	0
0	1	0
1	0	0
1	1	1

[CO1, BTL2, 01 marks]

Q2. For a seven-variable logic function, what is the total number of rows in the truth table?

Ans:

$$2^7 = 128$$

[CO1, BTL1, 01 marks]

Q3. What is the dual of the function X , if $X = (A' + B + C)(C + 0)$?

Ans:

$$X_D = (\bar{A} \cdot B \cdot C) + (C \cdot 1)$$

[CO1, BTL1, 01 marks]

Q4. What is the minimum number of gates required to implement a 2-input NOR gate using only 2-input NAND gates?

Ans:

$$4 \quad (\frac{1}{2} M)$$

[CO1, BTL4, 01 marks]

Q5. For the function given f , what is the value of f for inputs

(i.) $(A, B, C) = (1, 1, 0)$ and (ii.) $(A, B, C) = (0, 0, 0)$

$$f = AB' + BC' + CA$$

[CO1, BTL2, 01 marks]

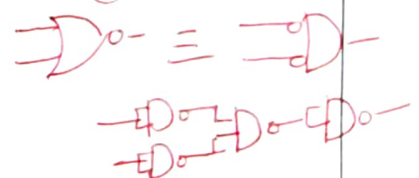
Ans:

(i.) 1

(ii.) 0

ROUGH WORK

(1/2 M)



$$f = AB' + BC' + CA$$

$$= 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 = 1$$

$$f = 0 \cdot 1 + 0 \cdot 1 + 0 \cdot 0 = 0$$

Q6. Identify canonical SOP expression among the following: (choose the correct option) with reasoning. [CO1, BTL2, 01 marks]

- (a) $F(x,y,z) = xyz + x'y'z'$
 (b) $F(x,y,z) = x'y + x'z + xz'$
 (c) $F(x,y,z) = xyz + xy'z + xz$

(1/2 M)

Ans:

Each product term is minterm (1/2 M)

Q7. Write the minterm list if the maxterm list is $f(a, b, c, d) = \prod M(2, 5, 6)$.

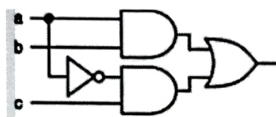
Ans:

[CO1, BTL2, 01 marks]

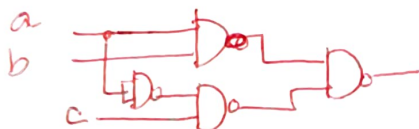
$\sum m(0, 1, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15)$

Q8. Convert the circuit to all NAND gates

[CO1, BTL1, 01 marks]



Ans:



Q9. Find the complement of function $F(a,b,c) = a'b' + ac + ac'$

Ans:

$$F' = \overline{a'b' + ac + ac'}$$

[CO1, BTL1, 01 marks]

$$F' = \overline{a'b'} \cdot \overline{ac} \cdot \overline{ac'}$$

$$= (a+b) \cdot (\bar{a} + \bar{c}) \cdot (\bar{a} + c)$$

Q10. Implement 2-input OR Gate using 2-input NAND Gates only.

Ans:

[CO1, BTL1, 01 marks]



Course Outcome (CO) / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	10	BTL1	05
CO2	00	BTL2	04
CO3	00	BTL3	00
CO4	00	BTL4	01
-	-	BTL5	00
-	-	BTL6	00

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Marks Obtained										
Total (out of 10)						Sign of student			Sign of faculty	

Q1. For the function given f , what is the value of f for inputs

(i.) $(A, B, C) = (1, 1, 0)$ and (ii.) $(A, B, C) = (0, 0, 0)$?

$$f = AB' + BC' + CA$$

[CO1, BTL2, 01 marks]

Ans: (i.) 1

(ii.) 0

Q2. For a eight-variable logic function, what is the total number of rows in the truth table?

[CO1, BTL1, 01 marks]

Ans:

$$2^8 = 256$$

Q3. What is the dual of the function X , if $X = (B+C)(A'+C+0)$?

Ans:

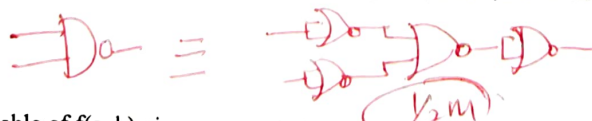
[CO1, BTL1, 01 marks]

$$X_D = (B \cdot C) \uparrow (\bar{A} \cdot C \cdot 1)$$

Q4. What is the minimum number of gates required to implement a 2-input NAND gate using only 2-input NOR gates?

Ans:

[CO1, BTL4, 01 marks]



Q5. Write Truth table of $f(a, b)$ given, output is low if "input is odd and greater than one".

Ans:

[CO1, BTL2, 01 marks]

a	b	f(a, b)
0	0	1
0	1	1
1	0	1
1	1	0

AB ROUGH WORK

1, 1, 0

$$f = A\bar{B} + B\bar{C} + CA$$

$$= 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 = 1$$

ABC
0 0 0

$$f = 0 \cdot 1 + 0 \cdot 1 + 0 \cdot 0 = 0$$

(4) No dia - 1/2 M.
(1/2 M)

Q6. Identify canonical SOP expression among the following: (choose the correct option) with reasoning. [CO1, BTL2, 01 marks]

- ✓ (a) $F(X, Y, Z) = XYZ + X'Y'Z'$ (1/2M)
 (b) $F(X, Y, Z) = X'Y + X'Z + XZ'$
 (c) $F(X, Y, Z) = XYZ + XY'Z + XZ$

Ans:

Each product term is minterm (1/2M)

Q7. Write the minterm list if the maxterm list is $f(a, b, c, d) = \prod M(2, 8, 10)$.

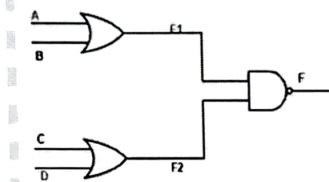
Ans:

[CO1, BTL2, 01 marks]

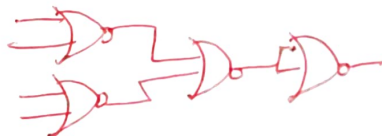
$\sum m(0, 1, 3, 4, 5, 6, 7, 9, 11, 12, 13, 14, 15)$

Q8. Convert the circuit to all NOR gates

[CO1, BTL1, 01 marks]



Ans:



Q9. Find the complement of function $F(A, B, C) = A'B' + AC' + AC$?

Ans:

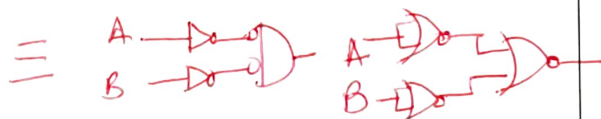
[CO1, BTL1, 01 marks]

$$F'(A, B, C) = \overline{A'B' + AC' + AC} \\ = \overline{A'B'} \cdot \overline{AC'} \cdot \overline{AC} = (A+B) \cdot (\bar{A}+C) \cdot (\bar{A}+C)$$

Q10. Implement 2-input AND Gate using 2-input NOR Gates only.

Ans:

[CO1, BTL1, 01 marks]



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